Where the Headroom Lies: Block ℓ_p-Norm Statistics for Cyclic Join Bounds

Extended version: full proofs and additional experiments

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This note accompanies the ICDE 2027 submission and gives the proofs that the paper states in condensed form: the ℓ_p Composition Inequality (the validity of every LpH-CEG bound) and the Cyclic Improvement theorem, and the $bs = 1$ experiments on JOB, a TPC-H-like schema, data size and skew. Notation follows the paper.

1 The ℓ_p Composition Inequality

Setup. A hyperpath $P = (S_0{=}\emptyset, F_1, S_1, \ldots, F_m, S_m)$ adds a block of relations F_j at step j, $S_j = S_{j-1} \cup F_j$, and S_m is the set of all relations. Write $A_j = \mathrm{attr}(F_j)$, $V_j = \mathrm{attr}(S_j)$ ($V_0 = \emptyset$, $V_m = V$), the interface $I_j = A_j \cap V_{j-1}$ and $E_j = A_j \setminus I_j$. Then

(F1) $E_j = V_j \setminus V_{j-1}$, so $\{E_j\}_j$ partitions V;

(F2) $I_j \subseteq V_{j-1}$.

Weights $w_{j,p} \geq 0$ ($p \in \mathcal{P} \subseteq [1, \infty]$) satisfy the attribute-wise cover constraint of the paper: for every attribute $a \in V$, bound at step $j(a)$ (i.e. $a \in E_{j(a)}$),

$$w_{j(a)} + \sum_{j' :\, a \in I_{j'}} s_{j'} \;\geq\; 1, \qquad w_j := \sum_p w_{j,p}, \quad s_j := \sum_p w_{j,p}/p \;\; (1/\infty = 0). \tag{C}$$

A *polymatroid* is $h : 2^V \to \mathbb{R}$ with $h(\emptyset) = 0$, monotone and submodular; $h(X \mid Y) := h(X \cup Y) - h(Y)$.

Lemma 1 (Conditioning)**.** *For a polymatroid h and $Y \subseteq Z$: $h(X \mid Y) \geq h(X \mid Z)$.*

Proof. Submodularity on $X \cup Y$ and Z gives $h(X \cup Y) + h(Z) \geq h(X \cup Y \cup Z) + h((X \cup Y) \cap Z)$, and $h((X \cup Y) \cap Z) \geq h(Y)$ by monotonicity since $Y \subseteq (X \cup Y) \cap Z$. Rearrange. □

Lemma 2 (Fractional subadditivity)**.** *Let f be monotone submodular on subsets of a finite set with $f(\emptyset) = 0$, and let $\lambda_i \geq 0$ satisfy $\sum_{i:\, a \in T_i} \lambda_i \geq 1$ for every $a \in S$. Then $\sum_i \lambda_i f(T_i) \geq f(S)$.*

Proof. Order $S = (a_1, \ldots, a_k)$ and let $S_{<t} = \{a_1, \ldots, a_{t-1}\}$. By monotonicity, the chain rule and Lemma 1, $f(T_i) \geq f(T_i \cap S) = \sum_{a_t \in T_i} f(a_t \mid T_i \cap S_{<t}) \geq \sum_{a_t \in T_i} f(a_t \mid S_{<t})$. Weighting by λ_i and exchanging sums, $\sum_i \lambda_i f(T_i) \geq \sum_t f(a_t \mid S_{<t}) \sum_{i:\, a_t \in T_i} \lambda_i \geq \sum_t f(a_t \mid S_{<t}) = f(S)$, using $f(a_t \mid S_{<t}) \geq 0$. □

Lemma 3 (The obligation)**.** *For every polymatroid h and every w satisfying (C),*

$$\sum_j \Big[s_j\, h(I_j) + w_j\, h(E_j \mid I_j) \Big] \;\geq\; h(V).$$

Proof. Let $g_j = h(E_j \mid V_{j-1})$, $\delta_j = \max(0, 1 - w_j) \in [0,1]$ and $f_j(S) = h(S \mid V_{j-1})$ for $S \subseteq E_j$; f_j is monotone, submodular and $f_j(\emptyset) = 0$.

(i) For each j', the chain rule along the partition (F1) and Lemma 1 give $h(I_{j'}) = \sum_j h(I_{j'} \cap E_j \mid I_{j'} \cap V_{j-1}) \geq \sum_j f_j(I_{j'} \cap E_j)$. Hence $\sum_{j'} s_{j'} h(I_{j'}) \geq \sum_j \sum_{j'} s_{j'} f_j(I_{j'} \cap E_j)$.

(ii) Fix j. If $\delta_j > 0$ then by (C) every $a \in E_j$ has $\sum_{j' : a \in I_{j'}} s_{j'} \geq 1 - w_j = \delta_j$, so the sets $I_{j'} \cap E_j$ with weights $s_{j'}/\delta_j$ fractionally cover E_j, and Lemma 2 gives $\sum_{j'} s_{j'} f_j(I_{j'} \cap E_j) \geq \delta_j\, g_j$. If $\delta_j = 0$ the same inequality holds because the left side is non-negative.

(iii) By (F2) and Lemma 1, $h(E_j \mid I_j) \geq g_j \geq 0$, and $w_j \geq 1 - \delta_j$, so $w_j h(E_j \mid I_j) \geq (1 - \delta_j) g_j$.

Adding (i)–(iii): the left side is at least $\sum_j [\delta_j + (1 - \delta_j)] g_j = \sum_j h(E_j \mid V_{j-1}) = h(V)$ by the chain rule and (F1). □

Theorem 1 (ℓ_p Composition Inequality)**.** *Let D be a database, $\mathcal{Q}(D) \neq \emptyset$, and let $\mu_p(F_j \mid I_j)$ be the ℓ_p-norm of the degree sequence of E_j-values per I_j-value in a relation $R_j \supseteq \pi_{A_j} \mathcal{Q}(D)$—for instance the block join $\bowtie_{R \in F_j} R$ computed exactly on D. If w satisfies (C), then*

$$|\mathcal{Q}(D)| \;\leq\; \prod_{j=1}^{m} \prod_{p \in \mathcal{P}} \mu_p(F_j \mid I_j)^{\,w_{j,p}}.$$

Proof. Let h be the entropy of the uniform distribution on $\mathcal{Q}(D)$; it is a polymatroid and $h(V) = \log_2 |\mathcal{Q}(D)|$. The marginal on A_j is supported on $\pi_{A_j} \mathcal{Q}(D) \subseteq R_j$, so the ℓ_p degree inequality of Abo Khamis et al. (PODS 2024) gives, for $p \in [1, \infty]$, $\frac{1}{p} h(I_j) + h(E_j \mid I_j) \leq \log_2 \mu_p(F_j \mid I_j)$. Multiply by $w_{j,p} \geq 0$, sum, and apply Lemma 3. □

Scope. The proof uses only monotonicity and submodularity, so it holds over the whole polymatroid cone and for every hyperpath, block size and $p \geq 1$. It requires (C) exactly: an LP solver may return weights violating (C) by about 10^{-7}; scaling w by $1/\min_a(\text{coverage of } a)$ restores (C) and changes the bound by a factor $1 + O(10^{-7})$, which the released code does not apply. It requires exact statistics of the database being bounded (or deterministic upper bounds on them, since the bound is monotone in each μ); a sampled or sketched statistic, such as the Hutchinson estimate of the common-neighbour ℓ_2 in the paper, is outside the guarantee. When some block join is empty the query is empty and the inequality holds trivially (the released LP gives such a factor weight zero, which only loosens the bound).

Proposition 1 (Necessity)**.** *If nonnegative weights $(w_{j,p})$ make the product above an upper bound on $|\mathcal{Q}(D)|$ for every database D, they satisfy (C). Hence on a fixed hyperpath the cover LP is optimal over all weightings of its statistics that are valid on every database.*

Proof. Fix $a \in V$ and $N \geq 2$, and let D_N give a the domain $[N]$ and every other attribute a single value, each relation holding all tuples over these domains. Then $|\mathcal{Q}(D_N)| = N$, and the block join of F_j has N tuples if $a \in A_j$ and one otherwise, so $\mu_p(F_j \mid I_j)$ is $N^{1/p}$ if $a \in I_j$, N if $a \in E_j$, and 1 if $a \notin A_j$. The bound is N^{c_a} with c_a the left side of (C) at a, and $N^{c_a} \geq N$ forces $c_a \geq 1$. (The entropy of D_N's join is the modular polymatroid $h(S) = [a \in S] \log_2 N$, so the proposition also holds over the polymatroid cone.) □

Why the elementary argument does not suffice. If $w_j \geq 1$ for every step, Lemma 3 follows from (iii) alone. The LP's optima on cyclic queries have $w_j < 1$ (e.g. 1/3 on the closing step of a cycle); the deficit δ_j is paid for by later interfaces, which is exactly step (ii). The cyclic improvement and the need for Lemma 2 have the same source.

Independent checks. An LP over the polymatroid cone decides Lemma 3 at fixed weights; enumerating every vertex of the feasible weight region decides it for all weights. Over 6,197 hyperpath structures on eleven shapes with $|V| \leq 6$ and block sizes 1–3 the minimum slack is 0 to solver precision; an adversarial search over random abstract hyperpath schemes with $|V| \leq 5$ found no violation; and over 207,000 database–block-size checks the bound never fell below the exact count (`scripts/shannon_check_wide.py`, `scripts/validity_search.py`).

2 The Cyclic Improvement theorem

Order the cycle C_k as $R_1(a_1, a_2), \ldots, R_k(a_k, a_1)$ and extend in this order with single-relation blocks. Let $N_j = |R_j|$, M_j the maximum degree and $\mu_p(R_j)$ the ℓ_p-norm of the degree sequence of R_j from a_j to a_{j+1}.

Structure. Step 1 has interface $\emptyset$ and extension $\{a_1, a_2\}$: its degree sequence is the single value N_1. A middle step $2 \leq j \leq k-1$ has interface $\{a_j\}$, extension $\{a_{j+1}\}$ and $\mu_\infty = M_j$. The closing step k has interface $\{a_k, a_1\}$ and empty extension; its conditional degrees are all 1, so $\mu_\infty = 1$ and $\mu_1 = N_k$. With $p = \infty$ and unit weights this gives $B_{\max} = N_1 \prod_{j=2}^{k-1} M_j$.

A feasible point for every common $p \geq 1$. Put weight $\frac{p}{p+1}$ with exponent p on every middle step, $\frac{p}{p+1}$ on step 1 and $\frac{1}{p+1}$ on the closing step with exponent 1. The cover constraints read: a_1: $\frac{p}{p+1} + \frac{1}{p+1} = 1$; a_2: $\frac{p}{p+1} + \frac{p/(p+1)}{p} = 1$; a_i for $3 \leq i \leq k-1$: $\frac{p}{p+1} + \frac{1}{p+1} = 1$; a_k: $\frac{p}{p+1} + \frac{1}{p+1} = 1$. All are tight, so the LP optimum satisfies

$$B_{\mathrm{LP}} \leq N_1^{\frac{p}{p+1}} N_k^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \mu_p(R_j)^{\frac{p}{p+1}} \quad \text{and} \quad \frac{B_{\max}}{B_{\mathrm{LP}}} \geq \Big(\frac{N_1}{N_k}\Big)^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \frac{M_j}{\mu_p(R_j)^{p/(p+1)}}.$$

The exponent must be common (or non-increasing). The constraint of the attribute between middle edges j and $j{+}1$ is $\frac{p_j}{p_j+1} + \frac{1}{p_{j+1}+1} \geq 1$, i.e. $p_{j+1} \leq p_j$. Choosing each edge's best exponent independently is therefore not feasible in general (e.g. $p_j = 1$, $p_{j+1} = \infty$ covers that attribute only to $\frac{1}{2}$); the LP may still mix exponents within a step.

Growth. At $p = 2$ a middle factor $M_j/\mu_2(R_j)^{2/3}$ exceeds 1 exactly when $\mu_2(R_j)^2 < M_j^3$: the sum of squared degrees is below the cube of the maximum degree. A few high-degree values satisfy it; many low-degree ones need not (n values of uniform degree d qualify iff $n < d$, and 10^4 values of degree 1 plus one of degree 20 do not). When every middle factor is at least $c > 1$ the ratio grows geometrically in the number $k-2$ of middle edges. As $p \to \infty$ every factor tends to 1, and LpH-CEG returns $\min(B_{\max}, B_{\mathrm{LP}})$, so the realized ratio is always at least 1. The boundary factor $(N_1/N_k)^{1/(p+1)}$ does not depend on k; it is 1 for a uniform cycle.

Acyclic queries. On a star the cover constraints force unit weight on every step and the LP returns the max-degree bound. On a path the LP can use an interface at $p = 1$ in place of an extension (on a 4-path it returns weights $(1, 0, 1)$), so acyclicity alone does not force unit weights; what redundant coverage on cycles adds is the slack exploited above.

3 Additional $bs = 1$ experiments: JOB, TPC-H-like, data size and skew

On the JOB-medium subset (CI: 3K, MK: 2K, MC: 1K rows) the two co-actor 4-cycles improve 1.94× (with keywords) and 2.80× (with companies) at $bs = 1$ with weights w=0.5, and a star query by exactly 1.00×; the IMDb suite of the paper is the systematic version. On a small TPC-H-like synthetic schema (SF=0.01; not `dbgen` data) every cyclic query we built yields 1.00×; LpH-CEG needs *cyclic structure* together with *degree skew* in a middle relation. On synthetic cycle-4 (10 seeds) the improvement grows with data size, from 5.9±0.7× at 100 vertices to 33.9±1.5× at 2K (relative std <12%; a single-run sweep reaches 57× at 5K), and peaks at moderate skew: 2.3±0.5× at α=0.5, 16.1±0.8× at α=2.0, 6.6±0.4× at α=3.0. At low skew all ℓ_p-norms nearly coincide, at extreme skew ℓ_∞ is already near-tight, and in between ℓ_2 captures the heavy tail.

Every number in this section is produced by the scripts in the repository and checked by its claim audit.

4 Multi-way ordering by per-step residuals

The per-step residuals can order a worst-case-optimal join greedily: GenericJoin iterations drop 2.3× on the Hetionet cycle-4 in our simulator, but over 66 synthetic queries the bound-guided order is 1.3% worse at the median (20 wins, 43 losses; Wilcoxon p=0.003, indicative, as queries share shapes and seeds), so it is a targeted heuristic.

5 Complexity

Collecting $\mu_p(F \mid I)$ for a single relation is one scan, $O(N)$; for a two-relation block it materializes the pair join, except for the interfaces computable from single-relation degree tables (all but the common-neighbour interface of a binary block; paragraph "Which statistic needs the join" of the paper). The cover LP has $m|\mathcal{P}|$ variables and $|V|$ constraints; it solves in milliseconds for the queries in the paper. The flow LP over a $bs \leq 2$ catalog is polynomial in the number of constraints and solves in 0.03–4.4 s at $k = 3$–16 over $4k$ orders.